
Learning Mixtures of Anisotropic Gaussian Kernels with Transformers

Anonymous Author(s)

Affiliation

Address

email

1 Introduction

Abstract

Transformers with billions of parameters have achieved remarkable success across various domains, such as language modeling and computer vision. Existing theoretical works analyze the empirically observed phenomena, such as training dynamics, in-context learning (ICL) generalization, depth/width effects, scaling laws, only in isolation and in restricted settings (single-layer, linear regression, fixed width), leaving open whether they are manifestations of a single underlying mechanism. In this paper, we show how a Transformer can provably learn a Gaussian Process (GP) with covariance kernel κ over a function space $p_\kappa(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$ from noisy observations $f_j(x_i) + \epsilon$. We consider a kernel being mixture of anisotropic Gaussian kernels parametrised by a mixing measure θ and show how in the mean-field regime, where the number of heads $\rightarrow \infty$, training a Transformer recovers θ with provable approximation and generalization guarantees. Our findings prove the convergence of the unsupervised training dynamics of Transformers, the emergence of scaling laws in the considered setting, and the necessity of the regularization of θ for the generalization properties of the ICL. We evaluate our architecture on synthetic and real-world benchmarks to support our theoretical findings.

This paper makes the following contributions:

- **Learning distributions over functions.** We formulate the problem of learning a Gaussian distribution $\mathcal{GP}(0, \kappa)$ over a function space such that $p(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$, where κ is an integral anisotropic Gaussian kernel, and the function space $\mathcal{F}$ is the reproducing kernel Hilbert space $\mathcal{H}_\kappa$ associated with κ , from noisy observations as a regularized optimization problem.
- **Connection to in-context learning.** We draw a connection of this problem to the in-context learning in Transformers Models, discuss how it can explain the convergence to the optimal solution, the empirically observed scaling laws, the role of the positional encodings and the importance of the regularization for the generalization properties.
- **Architectural implications and empirical validation.** Guided by the optimization problem, we propose a minor modification of the Transformer architecture that enforces the structural assumptions of our framework. Experiments on both synthetic and real-world benchmarks demonstrate improvements in size-generalization, interpretability, and computational efficiency.

33 **2 Related Work**

34 Several works have showed that a fixed Transformer can implemet a gradient descent for a simple
35 class of functions such as linear regression, Bayesian inference, etc.

2.1 Problem Setup

We are given a domain $\mathcal{X}$ equipped with a known probability measure μ , a function space $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow \mathbb{R}$, endowed with probability measure $\nu(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$ where $\mathcal{H}_\kappa$ is the reproducing kernel Hilbert space of functions induced by a *unknown* kernel κ . We observe a dataset $\mathcal{D} = \{D^{(1)}, \dots, D^{(m)}\}$ consisting of m observations $D^{(j)}$ of functions $f_1, \dots, f_m \sim \nu(\mathcal{F})$ each evaluated at n points drawn from μ :

$$\mathbf{X} = \{X_1, \dots, X_n\} \stackrel{\text{i.i.d.}}{\sim} \mu(\mathcal{X}), \quad \epsilon_{j,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

$$\mathcal{D} = \left\{ \left\{ (X_i, f_j(X_i) + \epsilon_{j,i}) \right\}_{i=1}^n \right\}_{j=1}^m.$$

Our goal is to learn the measure ν over $\mathcal{F}$. Given ν , predicting a new function $f \sim \nu$ from noisy observations $\{(x_i, y_i)\}_{i=1}^n$, where $y_i = f(x_i) + \epsilon_i$ and $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, reduces to MAP estimation:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n |y_i - f(x_i)|^2 + \frac{1}{2} \|f\|_{\mathcal{H}_\kappa}^2$$

Connection to ICL in Transformers. The setup described above is closely related to the in-context learning in Transformers. Given a prompt $(x_1, y_1), \dots, (x_n, y_n)$, where x_i are positional encodings and y_i are tokens, the task is to predict the next token at position x_{n+1} . Each prompt can be viewed as a noisy observation of a latent function $f \sim \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$, and distinct prompts correspond to distinct draws from $\exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$. The Transformer thus implicitly learns the shared structure of the functions, that is, the kernel κ itself, and uses it to generalize within a new context.

3 Gaussian Measures on the Reproducing Kernel Hilbert Space

In this section, we formally define the Gaussian Process in the Hilbert space $\nu(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$ and discuss its properties for anisotropic Gaussian kernels on the domain $[0, 1]^D$. We also show how knowing the kernel κ and the observed noisy functions values $Y_i = f(X_i) + \epsilon_i$ for $f \sim \mu_\kappa, i \in [n]$, provides a provable generalization guarantees for the kernel ridge regression estimator $\hat{f}_n$.

Let $(\mathcal{X}, \mathcal{A}, P_X)$ be a probability space where $\mathcal{X}$ is the input domain. Let $\kappa: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a symmetric positive semi-definite (PSD) kernel satisfying the integrability condition: $\int_{\mathcal{X}} \kappa(x, x) dP_X(x) < \infty$. We work primarily in the Hilbert space $L^2(P_X)$ with inner product $\langle f, g \rangle_{L^2} = \int_{\mathcal{X}} f(x)g(x) dP_X(x)$.

Definition 3.1 (Integral Operator). The integral operator $T_\kappa: L^2(P_X) \rightarrow L^2(P_X)$ associated with kernel κ is defined by

$$(T_\kappa g)(x) = \int_{\mathcal{X}} \kappa(x, y) g(y) dP_X(y). \quad (1)$$

Theorem 3.2 (Spectral Decomposition). The operator T_κ is self-adjoint, positive, and trace-class. It admits a spectral decomposition $T_\kappa \phi_k = \mu_\kappa \phi_k, k \in \mathbb{N}$, where:

- (i) $\mu_1 \geq \mu_2 \geq \dots > 0$ are the positive eigenvalues (counted with multiplicity),
- (ii) $\{\phi_k\}_{k=1}^\infty$ is an orthonormal basis of $\overline{\text{Range}(T_\kappa)} \subseteq L^2(P_X)$,
- (iii) $\sum_{k=1}^\infty \mu_\kappa = \text{Tr}(T_\kappa) = \int_{\mathcal{X}} K(x, x) dP_X(x) < \infty$.

The kernel admits the Mercer representation $K(x, y) = \sum_{k=1}^\infty \mu_\kappa \phi_k(x) \phi_k(y)$, with convergence in $L^2(P_X \otimes P_X)$.

Definition 3.3 (RKHS via Spectral Representation). The reproducing kernel Hilbert space (RKHS) $\mathcal{H}_\kappa$ associated with κ is

$$\mathcal{H}_\kappa = \left\{ f = \sum_{k=1}^\infty \theta_k \phi_k : \|f\|_{\mathcal{H}_\kappa}^2 := \sum_{k=1}^\infty \frac{\theta_k^2}{\mu_\kappa} < \infty \right\}, \quad (2)$$

70 equipped with the inner product

$$\langle f, g \rangle_{\mathcal{H}_\kappa} = \sum_{k=1}^{\infty} \frac{\theta_k \eta_k}{\mu_\kappa}, \quad \text{where } f = \sum_k \theta_k \phi_k, \quad g = \sum_k \eta_k \phi_k. \quad (3)$$

71 **Proposition 3.4** (Operator Characterization). *The RKHS admits the equivalent characterization*
 72 $\mathcal{H}_\kappa = \text{Range}(T_\kappa^{1/2})$ *with norm* $\|f\|_{\mathcal{H}_\kappa} = \|T_\kappa^{-1/2} f\|_{L^2}$. *The inclusion* $\mathcal{H}_\kappa \hookrightarrow L^2(P_X)$ *is continuous*
 73 *and dense in* $\text{Range}(T_\kappa)$.

74 **Remark 3.5.** The space $\mathcal{H}_\kappa$ is “smaller” than $L^2(P_X)$: for $f \in \mathcal{H}_\kappa$, the coefficients $\theta_k = \langle f, \phi_k \rangle_{L^2}$
 75 must decay faster than $\sqrt{\mu_\kappa}$.

76 **Definition 3.6** (Gaussian Measure via Characteristic Functional). The *Gaussian measure* μ_κ on
 77 $L^2(P_X)$ is the unique Borel probability measure with characteristic functional

$$\hat{\mu}_K(g) := \int_{L^2(P_X)} e^{i\langle g, f \rangle_{L^2}} d\mu_\kappa(f) = \exp\left(-\frac{1}{2}\langle g, T_\kappa g \rangle_{L^2}\right) \quad (4)$$

78 for all $g \in L^2(P_X)$.

79 **Theorem 3.7** (Existence and Characterization). *The measure* μ_κ *exists, is unique, and satisfies the*
 80 *following:*

81 (i) **Karhunen-Loève representation:**

$$f \sim \mu_\kappa \iff f = \sum_{k=1}^{\infty} \sqrt{\mu_\kappa} \xi_k \phi_k, \quad \xi_k \stackrel{\text{iid}}{\sim} N(0, 1) \quad (5)$$

82 where the series converges in $L^2(\Omega; L^2(P_X))$.

83 (ii) **Mean and covariance:**

$$\mathbb{E}_{\mu_\kappa}[f] = 0, \quad (6)$$

$$\mathbb{E}_{\mu_\kappa}[\langle f, g \rangle_{L^2} \langle f, h \rangle_{L^2}] = \langle g, T_\kappa h \rangle_{L^2}, \quad \forall g, h \in L^2(P_X). \quad (7)$$

84 Equivalently, $\mathbb{E}_{\mu_\kappa}[f \otimes f] = T_\kappa$ as operators.

85 (iii) **Finite-dimensional distributions:** For any $x_1, \dots, x_n \in \mathcal{X}$,

$$(f(x_1), \dots, f(x_n))^\top \sim N(0, K_{\mathbf{X}\mathbf{X}}), \quad (8)$$

86 where $(K_{\mathbf{X}\mathbf{X}})_{ij} = K(x_i, x_j)$.

87 **Remark 3.8** (Gaussian Process Equivalence). The measure μ_κ is the law of the Gaussian process
 88 $f \sim \mathcal{GP}(0, K)$.

89 The following theorem establishes $\mathcal{H}_\kappa$ as the Cameron-Martin space of μ_κ , providing the fundamental
 90 link between the RKHS and the Gaussian measure.

91 **Theorem 3.9** (Cameron-Martin). *Let* μ_κ *be the Gaussian measure on* $L^2(P_X)$ *and* $\mathcal{H}_\kappa$ *its associated*
 92 *RKHS. Then:*

93 (i) **Null RKHS:** *Sample paths lie outside the RKHS almost surely:* $\mu_\kappa(\mathcal{H}_\kappa) = 0$.

94 (ii) **Admissible shifts:** For $h \in L^2(P_X)$, define the translated measure $\mu_\kappa^h(A) := \mu_\kappa(A - h)$.
 95 Then

$$\mu_\kappa^h \ll \mu_\kappa \iff h \in \mathcal{H}_\kappa. \quad (9)$$

96 That is, $\mathcal{H}_\kappa$ is exactly the set of admissible shifts preserving absolute continuity.

97 (iii) **Radon-Nikodym derivative:** For $h \in \mathcal{H}_\kappa$,

$$\frac{d\mu_\kappa^h}{d\mu_\kappa}(f) = \exp\left(\langle h, f \rangle_{\mathcal{H}_\kappa} - \frac{1}{2}\|h\|_{\mathcal{H}_\kappa}^2\right) \quad (10)$$

98 where the pairing $\langle h, f \rangle_{\mathcal{H}_\kappa}$ extends to $f \in L^2(P_X)$ via

$$\langle h, f \rangle_{\mathcal{H}_\kappa} = \sum_{k=1}^{\infty} \frac{\eta_k \theta_k}{\mu_\kappa}, \quad h = \sum_k \eta_k \phi_k, \quad f = \sum_k \theta_k \phi_k. \quad (11)$$

99 **Corollary 3.10.** *The RKHS norm $\|\cdot\|_{\mathcal{H}_\kappa}$ measures the “cost” of deterministic shifts under the*
 100 *Gaussian measure.*

101 The formal expression “ $p(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$ ” is made rigorous by the following theorem.

102 **Theorem 3.11** (Onsager-Machlup). *Let μ_κ be a Gaussian measure on a Banach space $\mathcal{B} \supseteq \mathcal{H}_\kappa$*
 103 *with Cameron-Martin space $\mathcal{H}_\kappa$. Let $\|\cdot\|_{\mathcal{B}}$ be a measurable norm satisfying appropriate regularity*
 104 *conditions.¹ Then for any $f, g \in \mathcal{H}_\kappa$:*

$$\lim_{\epsilon \rightarrow 0} \frac{\mu_\kappa(\{h \in \mathcal{B} : \|h - f\|_{\mathcal{B}} < \epsilon\})}{\mu_\kappa(\{h \in \mathcal{B} : \|h - g\|_{\mathcal{B}} < \epsilon\})} = \exp\left(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2 + \frac{1}{2}\|g\|_{\mathcal{H}_\kappa}^2\right) \quad (12)$$

105 **Corollary 3.12** (RKHS Norm as Log-Density). *Taking $g = 0$:*

$$\lim_{\epsilon \rightarrow 0} \frac{\mu_\kappa(B_\epsilon(f))}{\mu_\kappa(B_\epsilon(0))} = \exp\left(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2\right). \quad (13)$$

106 *This is the precise sense in which “ $p(f) \propto \exp(-\frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2)$.”*

107 **Remark 3.13.** The function $f \mapsto \frac{1}{2}\|f\|_{\mathcal{H}_\kappa}^2$ is called the *Onsager-Machlup functional* of μ_κ . It plays
 108 the role of a “negative log-density” for the Gaussian measure.

109 3.1 Connection to Bayesian Inference and Kernel Ridge Regression

110 The following theorem establishes the connection between the Bayesian inference and the kernel
 111 ridge regression estimator: knowing the kernel κ and the observed noisy functions values $\mathbf{Y} = \{Y_i =$
 112 $f(X_i) + \varepsilon_i\}_{i=1}^n$ for $f \sim \mu_\kappa, \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, how well can we estimate this unknown function f ?

113 **Theorem 3.14** (Posterior and MAP Estimator). *Suppose we observe*

$$Y_i = f(X_i) + \varepsilon_i, \quad i = 1, \dots, n, \quad (14)$$

114 *where $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, $X_i \stackrel{\text{iid}}{\sim} P_X$, and the prior is $f \sim \mu_\kappa$. Then:*

115 (i) *The posterior $\mu_\kappa^{(n)}(\cdot | \mathbf{Y}, \mathbf{X})$ is a Gaussian measure.*

116 (ii) *The posterior mean coincides with the kernel ridge regression estimator:*

$$\hat{f}_n = \operatorname{argmin}_{f \in \mathcal{H}_\kappa} \left\{ \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - f(X_i))^2 + \|f\|_{\mathcal{H}_\kappa}^2 \right\} \quad (15)$$

117 (iii) *In closed form:*

$$\hat{f}_n(\cdot) = \mathbf{k}(\cdot)^\top (\kappa_{\mathbf{X}\mathbf{X}} + \sigma^2 I)^{-1} \mathbf{Y}, \quad (16)$$

118 *where $\mathbf{k}(x) = (\kappa(x, X_1), \dots, \kappa(x, X_n))^\top$.*

119 (iv) *The generalization error of the kernel ridge regression estimator $\hat{f}_n$ is given by:*

$$\|f - \hat{f}_n\|_{\mathcal{H}_\kappa}^2 \leq \text{TODO} \quad (17)$$

120 **Remark 3.15.** The RKHS norm appears as the regularizer in the MAP estimator precisely because it
 121 is the Onsager-Machlup functional of the GP prior. This establishes the equivalence:

122 *GP regression with covariance $K \longleftrightarrow$ Kernel ridge regression with RKHS $\mathcal{H}_\kappa$.*

123 4 The Integral Anisotropic Gaussian Kernel

124 Here we introduce a family of kernels considered in this work: integral anisotropic Gaussian kernels.
 125 They cover a wide range of kernels, including the isotropic Gaussian kernel, the radial Gaussian
 126 kernel, the finite mixture of anisotropic Gaussian kernels, the Wishart mixing measure, the rank-one
 127 (ridge function) measure, and the integral of the anisotropic Gaussian kernel.

¹Specifically, $\|\cdot\|_{\mathcal{B}}$ should be measurable and the small ball probabilities $\mu_\kappa(B_\epsilon(0))$ should satisfy certain asymptotic conditions; see Zeitouni (1989).

Definition 4.1 (Integral Anisotropic Gaussian Kernel). Let $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$ be a finite non-negative Borel measure on the cone of positive semi-definite matrices. The *integral anisotropic Gaussian kernel* induced by θ is the function $K_\theta : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by

$$K_\theta(x, y) = \int_{A \in \mathcal{S}_+^d} K_A(x, y) d\theta(A) = \int_{A \in \mathcal{S}_+^d} \exp\left(-\frac{1}{2}(x - y)^\top A(x - y)\right) d\theta(A). \quad (18)$$

We call θ the *mixing measure* or *structure measure* of the kernel.

Given functions $f_1, \dots, f_m \sim \mu_{\kappa_\theta}$, where κ_θ is an integral anisotropic Gaussian kernel the MLE estimator of corresponding mixing measure θ is given by:

$$\hat{\theta}_m = \arg \max_{\theta \in \Theta} \left\{ \prod_{i=1}^m p(f_i \mid \kappa_\theta) \right\} \quad (19)$$

where $p(f_i \mid \kappa_\theta) = \exp(-\frac{1}{2}\|f_i\|_{\mathcal{H}_\kappa}^2)$. Depending on the θ^* we can get the error rate of its MLE estimator $\hat{\theta}_m$, and the simpler the θ^* is, the better the generalization guarantees are. The following notion defines the simplicity of the mixing measure θ^* as its effective dimension.

Definition 4.2 (Effective Dimension). The **effective dimension** of θ^* relative to the model Θ is:

$$D(\theta^*) := \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon, \Theta_{\theta^*}, d_K)}{\log(1/\epsilon)} \quad (20)$$

where Θ_{θ^*} is a local neighborhood of θ^* in Θ , and $N(\epsilon, \cdot, d_K)$ is the ϵ -covering number in the kernel metric $d_K(\theta_1, \theta_2) = \|K_{\theta_1} - K_{\theta_2}\|$.

Remark 4.3. The effective dimension $D(\theta^*)$ captures the *local complexity* of the model around θ^* . It can vary with θ^* : the simpler the θ^* is, the smaller the effective dimension is, the better the generalization guarantees are.

Theorem 4.4 (General Rate Theorem). Let $\theta^* \in \Theta \subseteq \mathcal{M}_+(\mathcal{S}_+^d)$. Suppose the model $\{\kappa_\theta : \theta \in \Theta\}$ has local metric entropy around θ^* satisfying:

$$\log N(\epsilon, \{\theta \in \Theta : d_K(\kappa_\theta, \kappa_{\theta^*}) \leq \delta\}, d_K) \leq D(\theta^*) \cdot \log\left(\frac{\delta}{\epsilon}\right) \quad (21)$$

for all $0 < \epsilon < \delta$ sufficiently small. Then:

$$d_K(\hat{\theta}_m, \theta^*) = O_p\left(m^{-\frac{1}{2+D(\theta^*)/s}}\right) \quad (22)$$

where s is the smoothness parameter if θ^* has a density, or $s = \infty$ (giving rate $m^{-1/2}$) in the parametric case.

More precisely:

$$d_K(\hat{\theta}_m, \theta^*) = O_p\left(m^{-\alpha(\theta^*)}\right), \quad \alpha(\theta^*) = \frac{1}{2 + D(\theta^*)} \quad (23)$$

when the density has regularity $s = 1$.

5 Making the Optimization Problem Tractable via Transformers

Even though, the Theorem 4.4 provides a provable estimation error rate guarantees for recovering the mixing measure θ^* , hence, the kernel κ_θ , from the functions $f_1, \dots, f_m \sim \mu_{\kappa_\theta}$, in practice we only observe the noisy functions values $Y = f_j(X) + \varepsilon_i$ for $f_j \sim \mu_\kappa, \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. So given observations $\mathcal{D} = \{(\mathbf{X}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^m$ our optimization problem turns into

$$\hat{\theta}_m = \arg \max_{\theta \in \Theta} p_\theta(\mathcal{D}) = \arg \min_{\theta \in \Theta} -\frac{1}{m} \sum_{i=1}^m \log p_\theta(\mathbf{y}^{(i)} \mid \mathbf{X}^{(i)}) \quad (24)$$

where

$$-\log p_\theta(\mathbf{y} \mid \mathbf{X}) = -\log \int \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{y} - f(\mathbf{X})\|^2\right) d\mu_{\kappa_\theta}(f)$$

As our prior is a Gaussian Process, the exact Laplace approximation yields a closed-form expression for the log-likelihood:

$$-\log p_\theta(\mathbf{y} \mid \mathbf{X}) = \frac{1}{2} \mathbf{y}^\top (\kappa_\theta(\mathbf{X}, \mathbf{X}) + \sigma^2 I)^{-1} \mathbf{y} + \frac{1}{2} \log \det(\kappa_\theta(\mathbf{X}, \mathbf{X}) + \sigma^2 I) + \frac{n}{2} \log(2\pi)$$

Even though this problem is convex, there are two main challenges in implementing it in practice via gradient descent:

- Parameter θ is infinite-dimensional.
- The matrix inversion $(\kappa_\theta(\mathbf{X}, \mathbf{X}) + \sigma^2 I)^{-1}$ and the gradient with respect to θ is computationally prohibitive for large n .

First, let us discuss how to avoid the matrix inversion.

5.1 Bilevel Optimization Via Masking

One way to avoid the matrix inversion is to turn the optimization problem (24) into a bilevel optimization problem via masking approach.

$$\frac{1}{2} \mathbf{y}^\top (\kappa_\theta + \sigma^2 I)^{-1} \mathbf{y} = \frac{1}{2\sigma^2} \|\mathbf{y} - \hat{f}_\theta(\mathbf{X})\|_2^2 + \frac{1}{2} \|f\|_{\kappa_\theta}^2$$

where via Theorem 3.14 we have

$$\hat{f}_\theta = \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{2\sigma^2} \|\mathbf{y} - f(\mathbf{X})\|_2^2 + \frac{1}{2} \|f\|_{\kappa_\theta}^2$$

Given the collection of datasets $\mathcal{D} = \{(\mathbf{X}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^m$ we can split each pair $(\mathbf{X}, \mathbf{y})$ into a nonoverlapping masked and unmasked halves: $(\mathbf{X}, \mathbf{y}) = (\mathbf{X}_C, \mathbf{y}_C) \sqcup (\mathbf{X}_Q, \mathbf{y}_Q)$.

Substituting this into Equation (24) we have

$$\hat{\theta}_\mathcal{D} = \min_{\theta \in \Theta} \frac{1}{m} \sum_{i=1}^m \frac{1}{2\sigma^2} \|\mathbf{y}_Q - \hat{f}_\theta(\mathbf{X}_Q)\|_2^2 + \frac{1}{2} \|\hat{f}_\theta\|_{\kappa_\theta}^2 + \frac{1}{2} \log \det(\kappa_\theta(\mathbf{X}_Q, \mathbf{X}_Q) + \sigma^2 I)$$

where $\hat{f}_\theta = \operatorname{argmin}_{f \in \mathcal{H}_\kappa} \frac{1}{2\sigma^2} \|\mathbf{y}_C - f(\mathbf{X}_C)\|_2^2 + \frac{1}{2} \|f\|_{\kappa_\theta}^2$

The inner problem has parameter θ fixed and find the MLE of the function $\hat{f}_\theta$ given the unmasked dataset $\mathbf{X}_C$ and $\mathbf{y}_C$. The outer problem then finds the parameter θ that yields the best prediction of $\hat{f}_\theta$ on the masked dataset $\mathbf{X}_Q$ and $\mathbf{y}_Q$. Both problems are convex on their own but not jointly convex. However, we can solve it by alternating gradient descent with a following guarantee:

Theorem 5.1. *[TODO: add theorem]*

The Inner Problem. For a dataset $\mathbf{X}_C, \mathbf{y}_C$ we have

$$\hat{f}_\theta = \operatorname{argmin}_{f \in \mathcal{H}_\kappa} \frac{1}{2\sigma^2} \|\mathbf{y}_C - f(\mathbf{X}_C)\|_2^2 + \frac{1}{2} \|f\|_{\kappa_\theta}^2$$

One gradient step of the inner problem is:

$$\begin{aligned} f^{(t+1)} &= f^{(t)} - \eta \nabla_f \left[\frac{1}{2\sigma^2} \|\mathbf{y}_C - f(\mathbf{X}_C)\|_2^2 + \frac{1}{2} \|f\|_{\mathcal{H}_{\kappa_\theta}}^2 \right] \Big|_{f=f^{(t)}} \\ &= f^{(t)} - \eta \left(-\frac{1}{\sigma^2} \sum_{i=1}^{n_C} (y_i - f^{(t)}(x_i)) \kappa_\theta(\cdot, x_i) + f^{(t)} \right) \\ &= (1 - \eta) f^{(t)} + \frac{\eta}{\sigma^2} \sum_{i=1}^{n_C} (y_i - f^{(t)}(x_i)) \kappa_\theta(\cdot, x_i) \end{aligned}$$

The Outer Problem.

